

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 - ANALYTICAL PROBLEM SOLVING

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO2 - Manipulation (BL3, BL4)	Assessment:	Assessment Tool 1 - Analytical Problem Solving
Weightage:	30% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	K. KAVIYA	Reg. No.:	19A521262
Section / Batch:	1	Date:	17/07/2026

Code of Conduct

I K. KAVIYA (19A521262) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: [Signature]

Instructions

- This test contains 80 Analytical problem-solving questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- Each student should write 2 questions. No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Linked Data Structure, Search Data Structure	List Operations, Binary Search	1	50
Stack Data Structure, Queue Data Structures	Stack Notations, Priority Queue	2	50
GRAND TOTAL:		100 Marks	

16	192521262	KAVIYA K	1. Consider a linked list representing the sales of different products in a store for a week. Each node of the linked list contains the product ID and the number of units sold on a specific day. Implement a linked list data structure and provide functions to: Insert new sales data for a given product on a specific day. Calculate the total sales for each product at the end of the week. Find the product with the highest sales and its corresponding day. Display the sales data for a specific product for the entire week. Use your own data for product. 2. A calculator application receives the following expression: $(8+2)*(6-4)/2$
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Rubrics for Calculation

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Understanding	20	Demonstrates complete and precise understanding; identifies all constraints and edge cases	Shows clear understanding with minor gaps	Partial understanding; misses some constraints	Misinterprets problem or lacks clarity
Data Structure Selection	20	Selects optimal data structure with strong justification and comparison	Selects appropriate structure with reasonable justification	Selects somewhat suitable structure with weak reasoning	Incorrect or no data structure selection
Algorithm Design	20	Designs efficient, logically sound, and well-structured algorithm	Algorithm is correct with minor inefficiencies	Algorithm is partially correct or lacks clarity	Algorithm is incorrect or missing
Accuracy of Solution	30	Error-free, well-structured, optimized code/solution	Minor errors but overall functional	Multiple errors; partially working	Non-functional or missing solution
Clarity	10	Exceptionally clear, well-organized, uses diagrams effectively	Clear and organized with minor issues	Somewhat organized but lacks clarity	Poorly organized and unclear

Faculty In-charge

Signature: _____

Date: _____

Course Coordinator

Signature: _____

Date: _____

Program Director

Signature: _____

Date: _____

CO2-AT1-Data Structure

1. Consider a linked list representing the sales of different products in store for a week. Each node of linked list contains p-ID & number of units sold on a specific day.

Introduction:

A store records the sales of different products for each day of week. A linked list can be used to dynamically store this sales information. Each node contains the product ID, day, ~~units~~ sold and pointer to the next node.

The main operators & required data:

- > Insert new sales data
- > Calculate total weekly sales for each product.
- > Find the product with the highest sales and its

Corresponding day.

- > Display the sales of a particular product for the entire week.

Problem Analysis:

For each sales record, we need three important pieces of information:

Data	Purpose
Product ID	Identifies the product
Day	Identifies the day of sale
Units Sold	Stores the number of units sold.

Example products:

- * P101 - Laptop
- * P102 - Keyboard
- * P103 - Mouse

Algorithm
Insertion:
1. Create
2. ...

Example weekly data:

Product	Day	Units Sold
P101	Monday	5
P102	Monday	8
P103	Monday	12
P101	Tuesday	7
P102	Tuesday	6
P103	Tuesday	10
P101	Wednesday	4
P102	Wednesday	9
P103	Wednesday	15
P101	Thursday	8
P102	Thursday	5
P103	Thursday	11

Data Structure Design

A node can be represented as:

```
+-----+-----+-----+-----+
| Product ID | Day | Units Sold | Next |
+-----+-----+-----+-----+
```

For example:

P101 = Monday → 5 → NULL

Multiple nodes are connected to form the linked list.

Algorithm

In section:

1. Create a new node.
2. Store product ID, day & units sold.
3. If the list is empty, make the new node the head.
4. Otherwise, traverse to the end.
5. Attach the new node to the last node.

Total Sales:

1. Select a product ID.
2. Traverse the entire linked list.
3. Whenever the product ID matches, add its units sold.
4. Display the total.

Highest Sales:

1. Traverse all nodes.
2. Compare the units sold in each node.
3. Keep track of the maximum value.
4. Store its product ID & day.
5. Display the result.

Display Product Sales:

1. Ask for the product ID.
2. Traverse the list.
3. Display every node having that product ID.
4. This gives the product's sales throughout the week.

Expected Result:

For the sample data:

- > P101: $5 + 7 + 4 + 8 = 24$ units
- > P102: $8 + 6 + 9 + 5 = 28$ units
- > P103: $12 + 10 + 15 + 11 = 48$ units

Therefore:

Highest weekly - selling product = P103 (Mouse), 48 units.

The highest single-day sale is:

P103 on Wednesday = 15 units.

Conclusion:

The linked list provides a flexible way to store weekly sales records because new sales data can be inserted dynamically. By traversing the list, we can calculate total sales, identify the highest-selling product and day, and display all sales records for a selected product. This demonstrates the practical application of linked lists in store sales management systems.

- 2- A calculator application receives the following
 $(8+2)*(6-4)/2$.

Introduction:

A calculator needs to evaluate an arithmetic expression according to the standard operator precedence rules. Parentheses are evaluated first, followed by multiplication and division from left to right.

Problem Analysis:

Given:

$$(8+2) * (6-4) / 2$$

Operations used:

- > + Addition
- > - Subtraction
- > * Multiplication
- > / Division

Step-by-Step Evaluation:

First evaluate the expressions inside parentheses:

$$8 + 2 = 10$$

and

$$6 - 4 = 2$$

The expression becomes:

$$10 * 2 / 2$$

Now perform multiplication:

$$10 * 2 = 20$$

Then division:

$$20 / 2 = 10$$

Final Result:

$$(8+2) * (6-4) / 2$$

$$= 10 * 2 / 2$$

$$= 20 / 2$$

$$= 10$$

$$\text{Answer} = 10$$

Conclusion:

The expression is solved using operator precedence.

Parentheses are evaluated first, followed by multiplication and division. Therefore, the calculator produces the final result 10.

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